

Verify-R · 引用支持核验

示例元信息

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Synthetic Research Team · 2024

This is an invented demonstration document, not a published paper.

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3 Method

Each generated statement is paired with a passage and checked for textual support.

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6 Limitations

Textual support does not establish that the source passage itself is factually correct.

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4 Evaluation

Unsupported statements are flagged for manual review rather than silently accepted.

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Definition 1. For a fixed finite set Q of statements and fixed textual-support scores $s(q)$ in $[0, 1]$, define $A_{\text{tau}} = \{q \text{ in } Q: s(q) \geq \text{tau}\}$, with tau in $[0, 1]$. Scores measure the teaching checker output, not factual truth.

中文导读：对固定的有限陈述集合 Q , 给定固定文本支持分数 $s(q) \in [0, 1]$, 定义通过集合 $A_{\tau} = \{q \in Q : s(q) \geq \tau\}$ 。

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Lemma 1. Under Definition 1, if $\tau_1 \leq \tau_2$ then $A_{\tau_2} \subset A_{\tau_1}$.

Proof: Definition 1 admits q exactly when $s(q)$ reaches the threshold; $s(q) \geq \tau_2$ implies $s(q) \geq \tau_1$.

中文导读：采用 Definition 1 的筛选集合时, $\tau_1 \leq \tau_2$ 蕴含 A_{τ_2} 包含于 A_{τ_1} 。

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Theorem 1. Under Definition 1 and $\tau_1 \leq \tau_2$, $|A(\tau_2)| \leq |A(\tau_1)|$. Proof: Lemma 1 and Definition 1 give $A(\tau_2) \subseteq A(\tau_1)$; finite set cardinality preserves inclusion. This proves only count monotonicity, not better factual correctness or score calibration.

中文导读：在陈述和分数固定时, 提高阈值不会增加通过的陈述数量, 即 $|A_{\tau 2}| \leq |A_{\tau 1}|$ 。

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Lemma 2. Under Definition 1 and $\tau_1 \leq \tau_2$, $A(\tau_1) \text{ minus } A(\tau_2) = \{q \text{ in } Q: \tau_1 \leq s(q) < \tau_2\}$. Proof: Definition 1 translates membership and nonmembership into the two score inequalities.

中文导读： A_{τ_1} 中未通过 τ_2 的陈述, 恰为分数处于 $[\tau_1, \tau_2)$ 的陈述。

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Corollary 1. $|A_{(\tau_1)}| - |A_{(\tau_2)}|$ equals the number of q with $\tau_1 \leq s(q) < \tau_2$. Proof: Lemma 1 gives nested finite sets, so their cardinality difference equals the set-difference cardinality. Lemma 2 identifies that difference with the score interval.

中文导读：提高阈值后减少的通过数量, 等于分数位于 $[\tau_1, \tau_2)$ 的陈述数量。

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Corollary 2. If no statement score lies in $[\tau_1, \tau_2)$, the two thresholds accept the same number of statements. Proof: Corollary 1 gives a cardinality difference of zero. No factual-truth claim is implied.

中文导读：若没有陈述分数位于 $[\tau_1, \tau_2)$ ，提高阈值不会改变通过数量。